

VIJAYAADHITHAN TAMILSELVAN

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Experience

AI/ML Computational Science Analyst

Feb 2025 – Present

Accenture — Bangalore, India

- Engineered the GenAI data processing engine for the "Honda Agent" Claims Intelligence Workspace, designing a concurrent LLM extraction pipeline that successfully secured a major enterprise contract
- Built a dynamic data ingestion layer using Pandas to normalize multi-format automotive datasets, integrating a tri-layered AI confidence scoring algorithm to highlight critical issues
- Redesigned and optimized an end-to-end claims-to-project matching pipeline on Google Cloud Platform, increasing average matching accuracy from 40% to over 80% (peaking at 93%) and achieving an 85% binning coverage
- Engineered a scalable, serverless workflow using Cloud Run and GCP Workflows, enabling fault-tolerant processing of over 10,000 claims per run and reducing batch processing time by 25%
- Developed and maintained a real-time project indexing workflow to synchronize BigQuery data with a Google Vector Index, slashing the number of missed projects from 500 per week to less than 10
- Implemented rule-based filters, negative keyword matching, and a rejection reason audit file to improve retrieval accuracy and make model exclusions explainable

Software Test Engineer

Oct 2024 – Feb 2025

Juspay — Bangalore, India

- Automated testing of UPI payment flows to verify transaction accuracy and system reliability
- Utilized histogram analysis of pixel intensity distribution to compare images, ensuring accurate detection of UI inconsistencies and visual discrepancies across platforms

Programmer Analyst

Dec 2020 – Jun 2022

Cognizant Technology Solutions — Chennai, India

- Mitigated data migration risks between Bitbucket and Databricks, ensuring 100% data integrity
- Developed and executed Python test scripts to validate the accuracy and consistency of large-scale product data migrations, maintaining high data quality standards

Projects

Full-Stack AI-Assisted Booking Application (DoorStep)

- Architected a unified full-stack platform, successfully deploying both a live web application and a native Android application powered by a shared Node.js and Express backend
- Leveraged AI coding agents to rapidly generate and orchestrate React and Android frontends, accelerating cross-platform deployment to focus heavily on the high-level system architecture
- Designed a robust PostgreSQL database schema using Drizzle ORM and Zod to enforce strict end-to-end type safety and data validation across the platform

AI-Powered Project Staffing Assistant

- Developed a multi-agent RAG system using the AI Refinery SDK and a FAISS vector index to match employees to projects based on conversational user queries
- Engineered an advanced retrieval pipeline incorporating a BAAI/bge-reranker-large model and an LLM-based context compressor to improve search precision and analysis quality
- Built a conversational UI with Streamlit and a Flask backend, implementing a persistent JSON-based state manager to support multi-turn interactions

Education

Uppsala University — Sweden

Sep 2024

Master Programme in Data Science – Machine Learning and Statistics

Anna University — India

Mar 2020

Bachelor of Engineering – Computer Science

Publication

Is Twitch Sustainable? Quantifying the Carbon Footprint of Twitch Streaming

2024

- Investigated the carbon emissions of Twitch.tv through controlled experiments and real-world analysis, addressing the critical research gap in quantifying its environmental impact

Skills

Programming & Databases: Python, MySQL, PostgreSQL

ML/AI & Cloud: Scikit-learn, TensorFlow, PyTorch, Vertex AI, FAISS, Google Vector Index, BigQuery

Tools: Google Cloud Platform (GCP), Git, Jira, Docker

Frameworks & Visualization: Flask, Streamlit, Tableau, Apache Superset